



CAKAP LAH!

Survive Malaysia by Speaking BM

FINAL BUILD BRIEF for Rui Jie

ONE-LINE PITCH

You do not learn Bahasa Melayu by choosing answers. You survive real Malaysian situations by speaking it.

Team: Doulou & Kinny • Nezriq (BD) + Rui Jie (Engineer) • FINAL VERSION: 4 Sep 2026 • Deadline: 6 Sep 2026, 11:59 PM MYT

Build philosophy: small game, open-ended speech, clear feedback, real Malaysian flavour. No 97-screen language-learning empire by Sunday.

USE THIS VERSION ONLY. This FINAL BUILD BRIEF supersedes the earlier CAKAP LAH! development/product briefs. Freeze the MVP here unless a technical blocker forces a change.

1. Product Definition

CAKAP LAH! is a voice-controlled Bahasa Melayu practice mini-game for people who understand some BM but lack confidence speaking it. The player enters short Malaysian scenarios, hears a prompt through Revolab TTS, answers freely by voice, and receives feedback after Revolab STT transcribes the response.

What makes this version different

Typical language app	CAKAP LAH!
Repeat a fixed sentence	Respond freely to a situation
Choose from multiple-choice answers	Speak as the game controller
Generic travel/course content	Malaysian workplace and daily-life situations
One correct sentence	Many acceptable ways to express the same intent
Static feedback	NPC reacts to what the player actually said
Textbook language only	Useful natural BM, with formal/casual context where appropriate

Primary target user

The first target is Revolab staff and other non-native or low-confidence BM speakers working in Malaysia. The hackathon MVP is deliberately internal-first: useful enough for colleagues, simple enough to demo, and extensible later.

Product principles

- Voice first: the player should speak, not tap through a quiz.
- Open-ended: do not require an exact memorised sentence.
- Context matters: score the meaning and task completion before grammar perfection.
- Natural Malaysia: situations should feel recognisable to people living or working here.
- Encouraging, not embarrassing: humour can roast the situation, never the learner.
- Fast rounds: each challenge should take roughly 20–45 seconds.

2. Hackathon Guardrails

Requirement	What CAKAP LAH! must do
Revolab STT + TTS	Use BOTH in the working application. Using only one does not qualify.
Working demo	Must run end-to-end. A slide-only or fake click-through is not enough.
RevoCall	Do not use RevoCall in any form.
Internal code	Do not fork/reuse another internal Revolab codebase.
Other tools	Other frameworks, cloud services, libraries and non-speech APIs are allowed.

Requirement	What CAKAP LAH! must do
Submission	Deck (~10 slides recommended), working demo, mandatory screen recording, and run instructions/link/credentials.
Deadline	Sunday, 6 September 2026, 11:59 PM MYT.
Judging	Business impact, technical feasibility, user experience, meaningful STT/TTS alignment.
Registration	CONFIRM immediately that “Doulou & Kinny” with both members was registered in #hacki-hacki-thon. Official registration closed 20 August 2026. If not confirmed, contact the organisers now.
Platform access	Participants sign up at platform.revolab.ai/login . Free hackathon credits are provided. Use any STT/TTS models actually available in the portal.
Intellectual property	All work produced for this hackathon — including code, designs and materials — is owned by Revolab.

2A. Go / No-Go Gate — Confirm Before Development Continues

These checks remove avoidable Sunday-night surprises. Do not assume any unchecked item is fine.

Check	Status / action
Team registration	☐ Confirm Doulou & Kinny + both member names are registered.
Revolab portal login	☐ Rui Jie can log in successfully.
Hackathon credits	☐ Credits are visible / usable.
STT access	☐ At least one working STT call succeeds.
TTS access	☐ At least one working TTS call succeeds.
Submission contacts	☐ Keep the four required email recipients ready from Section 22A.

3. MVP Scope — Build This First

Definition of done: a player can complete at least three open-ended BM scenarios in one session, hear all prompts/feedback via TTS, answer by microphone, see the STT transcript, get a score and useful feedback, earn XP, and move to the next challenge without developer intervention.

Must have	Nice later / do not block MVP
3–5 playable scenarios	Accounts / login
Mic recording + STT transcription	Leaderboard

TTS prompt + TTS feedback	Complex avatar animation
Open-ended intent evaluation	Perfect pronunciation scoring
Score + feedback + XP	Large curriculum
Retry and next challenge	Multiplayer
Visible error/retry state	Analytics dashboard

MVP game loop

1. Load scenario
2. TTS sets the situation
- 3. Player speaks freely**
4. STT returns transcript
- 5. Evaluate meaning + language**
- 6. NPC/game responds**
7. TTS speaks feedback
8. Score + XP
9. Branch / next round

4. Technical Architecture

Layer	Responsibility	Minimum recommendation
Front end	Scenario card, record button, transcript, score, NPC reply, XP	Simple responsive web app
Audio capture	Record player response and stop after timeout/user action	Browser microphone recording
Revolab STT	Convert player audio into BM text	Use a portal-available STT model Rui Jie confirms works best
Evaluation layer	Judge intent, required semantic meaning, comprehensibility, naturalness and basic language quality	Rule-based + optional LLM for semantic evaluation
Branching engine	Choose NPC response / result state	Scenario JSON + deterministic rules
Revolab TTS	Speak scenario prompts and feedback	Use a portal-available TTS voice Rui Jie confirms
Session + conversation state	Score, XP, challenge number, attempts, current turn and prior dialogue needed for context	In-memory/local state is enough for the hackathon; pass relevant turn history to the evaluator
Deployment	Make judges able to try it	Simple hosted web app or clear local run instructions

Layer	Responsibility	Minimum recommendation
Credential security	Protect Revolab/API secrets	Never expose a secret API key in browser JavaScript. If the API uses a secret, call it through a small server-side proxy/backend.

4A. Developer Pre-Flight — Do This Before Building the UI

Goal: prove the speech loop first. A pretty interface with a broken speech pipeline scores nothing.

Item	Rui Jie must confirm / record
Portal + auth	Login works; API key/token/auth method confirmed from Revolab portal/docs.
STT model	Test Malay STT. Also test Manglish STT if exposed in the portal. Record the exact model ID/name chosen.
TTS model	Test Malay TTS. Also test Manglish TTS if exposed. Record the exact voice/model ID/name chosen.
Audio contract	Accepted input format, sample rate, file-size/length limit and timeout confirmed from current API docs. Do not invent these values.
STT proof	Speak one natural BM sentence → receive correct transcript.
TTS proof	Send one casual BM sentence → receive playable audio.
End-to-end proof	Mic → STT → evaluator → NPC reply → TTS works for one mission before UI polish.
Secret handling	If authentication uses a secret, keep it server-side; no keys committed to repo or exposed in front-end source.
Browser	Target demo browser chosen and microphone permissions tested.
Fallback	Local/hosted run method and backup machine/browser identified.
MODEL NOTE: Revolab's July 2026 Corporate Profile lists Malay STT/TTS and Manglish STT/TTS among its localized speech models. Rui Jie must still confirm the exact models currently exposed in the hackathon portal and their current API contract.	

Important: STT is not the same as a phoneme-level pronunciation engine.

For the hackathon, do not claim that CAKAP LAH! can measure pronunciation perfectly from STT alone. It can score what was recognised, whether the intended meaning was achieved, key BM concepts, sentence quality, and naturalness. If the STT repeatedly mishears an intended word, the game can suggest speaking that word more clearly, but phrase this as recognition feedback rather than scientific pronunciation grading.

SCORE DISCLAIMER: The 0–100 result is game feedback for communicative practice. It is NOT a certified Bahasa Melayu proficiency score, clinical/educational assessment, or phoneme-level pronunciation grade.

5. Open-Ended Answer Evaluation

The player response is intentionally random/open-ended. The evaluator must accept multiple valid phrasings that satisfy the situation.

Dimension	Weight	What to check
Intent / task completion	40%	Did the player accomplish what the scenario asked?
Required meaning / semantic concepts	25%	Did the response convey the needed ideas, regardless of exact wording? Example: “jangan letak banyak gula” can satisfy the same meaning as “kurang manis”.
Comprehensibility	20%	Is the recognised BM understandable as a coherent response?
Naturalness / grammar	15%	Is the wording reasonably natural and grammatically usable for the learner level?

Suggested score bands

Score	Game result	Behaviour
90–100	Power! / Critical Hit	Strong answer. Give concise praise + optional better phrasing.
75–89	Mission Passed	Meaning works. Mention one small improvement.
55–74	Almost there	Recognise what worked, give one correction, allow retry.
<55	Retry	Explain the missing intent/concepts. Do not shame the player.
No transcript	No score	Ask player to retry; likely silence/audio/STT issue.

5A. Language Policy — BM, Manglish & English

CAKAP LAH! teaches communicative confidence, so code-switching is handled by level rather than treated as an automatic failure.

Level	What is accepted	How to coach it
Beginner	BM with some English/Manglish can	Do not fail the mission. Show the BM

Level	What is accepted	How to coach it
	pass if the task is completed.	replacement: e.g. “less sweet” → “kurang manis”.
Intermediate	Code-switching can still pass, but reduce naturalness/language score slightly when a normal BM alternative exists.	Give one concise replacement or better phrasing.
Advanced	Expect predominantly BM, except natural proper nouns/technical terms.	Coach toward fluent, situation-appropriate BM.
NON-NEGOTIABLE: No exact-sentence matching. Score meaning first. Different wording that completes the task must be able to pass.		

Evaluation object — recommended output

```
{
  "intent_pass": true,
  "intent_score": 92,
  "semantic_score": 88,
  "comprehensibility_score": 90,
  "naturalness_score": 82,
  "overall_score": 88,
  "result": "pass",
  "what_worked": "Meaning was clear and appropriate.",
  "improvement": "Try 'kurang manis' instead of 'less sweet'.",
  "npc_reply": "Baik boss, teh tarik satu kurang manis!",
  "branch": "success"
}
```

5B. Evaluator Contract — Rui Jie Should Implement This Shape

The evaluator may use rules, an LLM, or a hybrid. The application should treat the evaluator as a strict structured service, not free-form chat.

Evaluator INPUT	Required content
scenario_id / level	Which mission and learner difficulty.
scenario_context	The situation the learner is trying to handle.
npc_prompt	The exact current NPC question/prompt.
task_goal	What communicative outcome the learner needs to achieve.
expected_semantics	Meanings that should be present — concepts, not literal keywords.
allowed_code_switch	Beginner / intermediate / advanced policy.
stt_transcript	What Revolab STT heard from this turn.

Evaluator INPUT	Required content
conversation_history	Relevant prior NPC/player turns. REQUIRED for multi-turn missions.
Evaluator OUTPUT — valid JSON only	Type / rule
intent_pass	boolean
intent_score	0–100 integer
semantic_score	0–100 integer
comprehensibility_score	0–100 integer
naturalness_score	0–100 integer
overall_score	0–100 integer; weighted by game scoring policy
result	success partial retry
what_worked	One short learner-friendly sentence
improvement	One short actionable coaching sentence
npc_reply	Context-aware response for Revolab TTS
branch	Scenario branch ID, e.g. success / clarify / retry
MALFORMED OUTPUT FALLBACK: Validate the JSON. If invalid, retry the evaluator once with a strict “JSON only” instruction. If it still fails, use a deterministic fallback branch + generic coaching message. Do not punish the learner for an evaluator/system error.	
MULTI-TURN RULE: “Boleh, sebelum pukul lima saya hantar” only makes sense if the evaluator knows the NPC just asked when the report will be ready. Always pass the minimum relevant conversation history.	

6. Scenario Data Structure

Keep scenarios data-driven so Nezriq can add/edit content without changing application logic.

Field	Purpose	Example
id	Unique challenge ID	mamak_01
level	Difficulty / progression	1
title	Game-facing name	Mamak Mission
context	Situation shown to player	You want a teh tarik with less sugar.
tts_prompt	What the NPC says aloud	Pesan minuman anda dalam Bahasa Melayu.
task	What the player must accomplish	Order one teh tarik, less sweet.
required_intents	Semantic task checks	order_drink, less_sweet

key_concepts	Useful lexical signals	teh tarik, kurang manis
sample_answers	Examples only, never exact-match list	Saya nak teh tarik satu, kurang manis.
success_replies	Possible NPC branches	Baik boss!
retry_hint	Helpful prompt after low score	Cuba sebut minuman dan cara anda mahu ia dibuat.
xp	Reward	100

7. Three Complete Playable Scenarios

SCENARIO 1 — MAMAK MISSION

Goal: order a drink naturally and handle a wrong order. Level: beginner. Tone: casual Malaysian BM.

Step	NPC / System	Player example	Evaluation / branch
1	TTS: “Anda di mamak. Pesan satu teh tarik kurang manis.”	“Boss, teh tarik satu, kurang manis.”	Intent pass: order + drink + less sweet.
2	Game: 95/100. NPC TTS: “Baik boss, teh tarik satu kurang manis!”	—	Award 100 XP.
3	TTS: “Eh, saya bagi Milo ais. Beritahu saya minuman ini salah.”	“Maaf boss, saya pesan teh tarik kurang manis tadi.”	Intent pass: polite correction + expected drink.
4	NPC TTS: “Aduh, salah meja. Saya tukar sekarang.”	—	Mission complete. Badge: Can Survive at Mamak.

Possible open-ended valid answers:

- “Saya nak teh tarik satu, kurang manis.”
- “Boleh bagi satu teh tarik kurang manis?”
- “Teh tarik satu boss, jangan manis sangat.”

SCENARIO 2 — OFFICE PANIC

Goal: respond to a colleague/manager about an unfinished report. Level: beginner–intermediate. Tone: workplace casual.

Step	NPC / System	Player example	Evaluation / branch
1	TTS: “Wei, report semalam dah siap ke?”	“Belum siap lagi. Saya tengah buat sekarang. Petang ni saya hantar.”	Intent: not finished + progress + commitment.
2	If strong answer, NPC TTS: “Okay, petang ni ya. Jangan esok pula.”	—	Mission passed +150 XP.
3	If vague answer, NPC TTS: “Petang ni boleh siap atau tak?”	“Boleh. Sebelum pukul lima saya hantar.”	Branch asks for concrete commitment.

4	Game result	—	Meaning score, natural BM feedback, boss patience meter joke.
---	-------------	---	---

Humour option: “HONESTY +10. CAREER PROSPECTS -20.” Use only for obviously comic demo mode, not as learner judgement.

SCENARIO 3 — CLIENT COUNTER CHALLENGE

Goal: explain politely that a requested document is incomplete and ask the client to provide one missing item. Level: intermediate. Tone: professional BM.

Step	NPC / System	Player example	Evaluation / branch
1	TTS: “Pelanggan bertanya, ‘Kenapa permohonan saya belum boleh diproses?’ Jawab dengan sopan.”	“Maaf, permohonan belum boleh diproses sebab dokumen masih tak lengkap.”	Intent: explain delay + incomplete documents + polite tone.
2	NPC TTS: “Dokumen apa yang masih diperlukan?”	“Kami masih perlukan salinan kad pengenalan. Lepas terima, kami boleh teruskan proses.”	Intent: specify missing item + next step.
3	NPC TTS: “Baik, saya akan hantar hari ini.”	—	Mission passed.
4	Game feedback	—	Highlight professional wording; suggest “belum lengkap” over overly casual alternatives.

8. UI Screens & States

Screen / state	What must be visible
Home	CAKAP LAH! logo/title, “Start Mission”, optional difficulty/level, XP
Scenario intro	Scenario title, context, NPC prompt, Listen button/replay
Listening	Large record/stop control, timer, “Cakap sekarang...”
Processing	Clear STT/AI processing state; never look frozen
Transcript	What the system heard; allow retry if wrong/empty
Result	Score, intent result, concepts, one feedback point, XP
NPC reaction	Branching reply displayed + spoken by TTS
Mission complete	Total XP, badge/one-liner, Next Mission
Error	Human-readable error + Retry; no dead end

Visual tone

Think “mini voice game” before “corporate training portal”: bold mission cards, progress/XP, readable transcript, one obvious microphone action. Avoid visual clutter. The live demo must make the STT → evaluation → TTS loop instantly obvious.

9. Branching Logic — Plain-English Pseudologic

1. Load scenario and current player state.
2. Play the scenario prompt using Revolab TTS.
3. Wait for microphone recording. Cap the recording duration.
4. Send audio to Revolab STT.
5. If no transcript: show “Tak dengar tadi — cuba lagi” and do not deduct points.
6. Normalise transcript for scoring (case/punctuation/filler handling).
7. Evaluate semantic intent, required meaning / semantic concepts, comprehensibility and language quality.
8. Generate structured evaluation output and choose success / partial / retry branch.
9. Select an NPC reply that matches that branch and the content of the answer.
10. Display transcript + score + one useful coaching note.
11. Play NPC/feedback reply using Revolab TTS.
12. Award XP, update session, and load next scenario.

10. Failure States, Edge Cases & Data

Case	Expected behaviour
Silence / mic not captured	Retry without zero score.
STT returns nonsense	Show transcript and offer Retry. Do not punish obvious system error.
Mixed BM-English	Accept at beginner levels if task is completed; coach toward more BM.
Alternative valid phrasing	Semantic evaluator should pass it even if sample answer differs.
Very long response	Trim/cap duration; evaluate core response.
Profanity / nonsense	NPC can respond lightly; score based on task completion.
STT/TTS API error	Visible retry state; cache scenario text so app remains understandable.
Network slow	Processing indicator + timeout/retry.
LLM/evaluator invalid JSON	Retry once with strict JSON output. If it still fails, use deterministic scenario-meaning/intent rules; do not fall back to literal keyword matching.
TTS unavailable	Display the text reply and Retry TTS; app should not crash.

Privacy for hackathon

Use only the minimum audio/transcript data needed for the session. Do not build employee performance monitoring into the MVP. If logs are retained for debugging, keep them limited and internal. The product pitch is language practice, not staff evaluation.

11. Rui Jie — Developer Task Breakdown

Priority	Task	Acceptance check
P0	Confirm platform access, STT and TTS model/API calls	One STT request + one TTS request works independently.
P0	Mic capture → STT transcript	Speak BM and transcript appears.
P0	TTS prompt playback	Scenario prompt is spoken.
P0	Scenario JSON/data loader	Can change scenario content without editing core logic.
P0	Evaluation function	Returns structured score + branch for open-ended answer.
P0	TTS feedback / NPC reaction	Generated/selected reply is spoken.
P0	XP + next mission	Player can complete 3 rounds.
P1	Retry / API failure states	No dead ends during demo.
P1	Basic game polish	Score card, XP, mission title, transcript are clear.
P2	Badges / extra humour	Only after core loop is stable.

12. Nezriq — Product / BD / Content Tasks

BM CONTENT QA OWNER: Nezriq (or another fluent natural-BM reviewer) must approve every demo prompt, acceptable semantic goal, coaching line and NPC response. We cannot ship a BM-learning game that teaches awkward BM. 😊

- Freeze the product story: “Survive Malaysia by Speaking BM.”
- Write and QA the first 5–8 scenarios with natural Malaysian BM.
- Define target user and business/internal value.
- Provide sample valid responses and intended semantic goals for each challenge.
- Write humour/feedback bank that is encouraging and workplace-safe.
- Test scenarios as a real low-pressure learner — do random answers, not scripted answers.
- Prepare the ~10-slide hackathon deck and live demo narration.
- Record/coordinate the mandatory screen recording and submission email.

13. Minimum Content Library

#	Mission	Task	Difficulty
1	Mamak Mission	Order a drink less sweet	Beginner

2	Wrong Order	Politely correct a wrong drink/food	Beginner
3	Dah Makan?	Respond naturally to a colleague	Beginner
4	Office Panic	Explain unfinished work + commitment	Beginner–Intermediate
5	Meeting Move	Ask to reschedule a meeting politely	Intermediate
6	Client Counter	Explain missing documents + next step	Intermediate
7	Direction Rescue	Ask/give simple directions	Beginner–Intermediate
8	Boss Level: Boleh Settle Hari Ni?	Negotiate deadline or confirm delivery naturally	Intermediate

14. Test Plan

Test	What to try	Pass condition
Happy path	Natural correct BM response	Correct intent branch, sensible score, TTS feedback.
Different valid wording	Use a phrasing not in sample answers	Still passes if meaning is correct.
Mixed language	BM + English answer	Beginner mission can pass with coaching.
Wrong intent	Answer unrelated to task	Low score / retry branch.
Silence	Say nothing	Retry, no zero/punishment.
Fast speech	Speak quickly	Transcript still handled; visible retry if not.
Noisy room	Moderate background noise	App fails gracefully if STT is poor.
API failure	Simulate/force error	Retry state, no crash.
TTS failure	Disable/fail TTS	Text reply remains visible.
Three-round session	Complete 3 missions consecutively	XP/state update and no reset bug.
Evaluator malformed JSON	Force/return invalid evaluator output	App retries once, then uses deterministic fallback without penalising learner.
Multi-turn context	Give a short answer that depends on prior NPC turn	Evaluator interprets it correctly using conversation history.

Test	What to try	Pass condition
Secret exposure check	Inspect browser source/network/repo	No secret API key is present in client bundle or committed source.
Demo browser/mic	Fresh browser session on demo machine	Mic permission, recording, playback and TTS all work.

15. Build Sequence & Deadline Plan

When	Rui Jie	Nezriq
4 Sep — First block	Confirm STT/TTS access. Build mic → STT transcript.	Freeze product name, game loop, first 3 scenarios.
4 Sep — Second block	Add TTS prompt and one-scenario evaluator.	Prepare scenario intents, sample answers, feedback.
4 Sep — End of day	One complete end-to-end mission must work.	Test random responses; log issues.
5 Sep — Morning	Add branching, XP, 3–5 missions, error states.	Complete deck content + demo script.
5 Sep — Afternoon	Polish only after stability.	Run structured UAT and random-answer testing.
5 Sep — Night	Freeze demo build / backup copy.	Record first screen recording.
6 Sep — Morning	Fix only blocking defects.	Final deck and submission package.
6 Sep — Target 8 PM	No new features. Demo-ready build.	Submit early; official deadline 11:59 PM MYT.

16. Acceptance Criteria — Definition of Done

FINAL GATE: “Done” means speech works, open-ended alternatives pass correctly, keys are protected, the demo survives retries, and the submission package is ready — not merely that the UI loads.

- ☐ Registration status has been confirmed.
- ☐ Selected STT/TTS model names/IDs and current API contract are recorded.
- ☐ Secret API credentials are not exposed client-side or committed to source.
- ☐ At least three missions work end-to-end with multi-turn context where needed.
- ☐ Two different valid phrasings for the same task can both pass.
- ☐ BM content for the live demo has been reviewed for natural Malaysian usage.
- ☐ Player hears a BM prompt through Revolab TTS.
- ☐ Player answers naturally through microphone without selecting a fixed answer.
- ☐ Revolab STT transcript is visibly shown.
- ☐ App produces a score and one useful coaching point.
- ☐ NPC/system response changes based on success / partial / retry logic.

- ☐ NPC/feedback response is spoken through Revolab TTS.
- ☐ XP or progress updates after a successful round.
- ☐ At least three scenarios work consecutively.
- ☐ Silence/API failure produces a Retry state, not a crash.
- ☐ Screen recording demonstrates the actual working app end-to-end.

17. Live Demo Script — 90 Seconds

Time	What happens	What judges should notice
0:00	Nezriq: “You don’t learn BM by choosing answers. You survive by speaking it.”	Clear product hook.
0:10	Start Mamak Mission. TTS gives situation.	Revolab TTS is integral.
0:20	Nezriq/Rui Jie gives an unscripted but valid BM answer.	Open-ended voice control.
0:30	Show STT transcript and evaluation.	Revolab STT + semantic evaluation.
0:40	NPC reacts and TTS speaks feedback.	Branching loop, not an API test.
0:55	Jump to Office Panic and answer differently.	Different scenario + random answer.
1:10	Show XP/badge and next mission.	Game mechanics.
1:20	Close: “Real Malaysian situations. Real speech. No multiple choice.”	Memorable differentiation.

Optional opening joke: “We realised some Revolites’ Bahasa Melayu is a production incident. So Rui Jie built monitoring.”

18. Recommended 10-Slide Deck

Slide	Title	Main message
1	CAKAP LAH!	Survive Malaysia by Speaking BM — Doulou & Kinny.
2	The Problem	People may understand BM but avoid speaking because practice feels awkward.
3	Why Existing Practice Feels Flat	Fixed answers and textbook phrases do not simulate real Malaysian conversations.
4	Our Solution	Voice-controlled mini-game with open-ended Malaysian scenarios.
5	How It Works	TTS → speak freely → STT → evaluate → NPC reacts → TTS feedback.
6	Why Revolab Speech Matters	STT and TTS are the core game controls,

		not add-ons.
7	Live Demo	Mamak Mission / Office Panic.
8	User & Business Impact	Confidence, onboarding, local communication, internal learning.
9	Technical Build	Architecture, evaluation logic, feasibility.
10	What We Built / What's Next	More scenarios, adaptive levels, departments/industries, analytics later.

19. Judging Alignment

Criterion	Our case
Business impact	Internal-first for Revolab staff BM confidence; clear expansion to HR/L&D, onboarding, BPO/customer service, hospitality, public-facing teams and Malaysian employers with international/non-BM-speaking staff.
Technical feasibility	Small web app, finite scenario data, existing Revolab STT/TTS, simple session state and branching.
User experience	Fast rounds, voice as the primary control, feedback immediately after speaking, recognisable Malaysian situations.
Alignment to objective	Both Revolab STT and TTS are mandatory to the game loop and demonstrated end-to-end.

19A. Commercial Expansion & Success Metrics

The hackathon MVP is internal-first. The commercial story is broader: organisations can use scenario packs to practise the exact Bahasa Melayu their people need at work.

Potential buyer / team	Relevant use
HR / Learning & Development	BM speaking practice for onboarding and workforce development.
BPO / Customer Service	Service-language scenarios, complaints, clarifications and customer handling.
Hospitality / Retail	Frontline Malaysian customer interactions.
International staff onboarding	Practical everyday/workplace BM instead of textbook-only learning.
Public-facing / government-facing teams	Professional BM scenario packs for counters, meetings and stakeholder interactions.

Success metrics

Learner metrics	Organisation-level metrics
Missions completed	Participation / active users
Attempts per mission	Mission completion rate
Average game score by scenario	Practice frequency
BM vs code-switch usage trend	Average score movement over repeated practice
Repeated-mission improvement	Most-practised / most-difficult scenario categories
Interpret scores as gameplay/training indicators only — not a formal BM proficiency certification.	

20. Doulou & Kinny Humour Guide

Use	Examples
Mission names	Mamak Mission; Office Panic; Boss Level
Success	“Malaysia has accepted your application.” / “Dah boleh cakap.”
Retry	“Bahasa Melayu belum give up on you. Cuba lagi.”
NPC humour	“Okay, petang ni ya. Jangan esok pula.”
Game stats	Boss patience remaining: 43%
Avoid	Jokes about ethnicity, nationality, intelligence, accent, or a learner being stupid.

Rule: the joke is always about the situation/game. The learner should feel brave enough to try again.

21. Post-Hackathon Roadmap

Phase	Possible extension
Next	More scenarios, difficulty levels, adaptive hints, saved progress.
Learning	Formal vs natural BM explanations, vocabulary review from previous mistakes.
Workplace packs	Sales BM, customer service BM, government-facing BM, technical/site BM.
Game layer	Streaks, leaderboards (opt-in), team challenges, seasonal missions.
Speech	If available later, add dedicated pronunciation/phoneme scoring rather than inferring it from STT.

Analytics	Learner-owned progress; aggregated training insights without turning it into staff surveillance.
External product	Localisation for organisations onboarding non-BM speakers in Malaysia.

22. Submission Checklist

- ☐ Working application/demo
- ☐ Deck presentation (PDF recommended, around 10 slides)
- ☐ Mandatory screen recording
- ☐ Demo link / credentials / clear run instructions
- ☐ Email subject: Revolites Hackathon - Doulou & Kinny
- ☐ Include team name, member names and one-line description
- ☐ Email to all required recipients listed in the hackathon brief
- ☐ Submit before Sunday, 6 September 2026, 11:59 PM MYT

22A. Exact Judge Submission Package

Send the submission to ALL four recipients before the deadline:

Recipient
shen@revolab.ai
daniel.gun@revolab.ai
cheelam@revolab.ai
dania.hassin@revolab.ai

Subject: Revolites Hackathon - Doulou & Kinny

READY-TO-COPY EMAIL

Hi all,

Please find our submission for the Revolites Hackathon 2026.

Team: Doulou & Kinny

Members: Nezriq and Rui Jie

Project: CAKAP LAH! — Survive Malaysia by Speaking BM

One-line description: CAKAP LAH! is a voice-controlled Bahasa Melayu mini-game where users survive real Malaysian situations by speaking naturally, with Revolab STT understanding their open-ended responses and Revolab TTS bringing the game characters to life.

Included:

- Deck presentation
- Demo link / credentials / run instructions
- Mandatory screen recording

Thank you.
Doulou & Kinny

22B. Demo Backup Plan — Murphy Is Also Invited

Risk	Backup / mitigation
Hosted demo fails	Keep a tested local run method on Rui Jie's machine.
Browser/mic issue	Use the browser already tested in the final test plan; keep a second browser ready.
Scenario content unavailable	Scenario JSON/content is bundled locally; no remote CMS dependency for MVP.
Evaluator transient error	One automatic retry, then deterministic fallback response.
STT/TTS/API outage during judging	Do not fake a live result. Explain the failure briefly and use the mandatory screen recording to prove the end-to-end working build.
Internet unstable	Have the completed screen recording, deck and local build already on the demo laptop.
TARGET: Final package ready by Sunday 8:00 PM MYT. 11:59 PM is the official deadline, not our personal upload strategy.	

BUILD THIS, NOT EVERYTHING

A judge should be able to hear a scenario, speak a completely natural BM answer, see what Revolab STT heard, watch the game understand the meaning, hear the NPC answer back through Revolab TTS, and immediately want to play the next round.

Source Notes

- Revolites Hackathon 2026 Requirements & Guidelines — source of registration, platform access, mandatory STT+TTS, prohibited RevoCall/internal code reuse, deliverables, judging criteria, recipients, deadline and IP ownership.
- Revolab Corporate Profile V3 Jul 2026 — lists localized Malay STT/TTS and Manglish STT/TTS models. The actual hackathon portal model IDs, endpoint/auth contract and limits must be confirmed from current portal/API documentation.

Doulou & Kinny target: rare enough to be remembered, small enough to actually work.